A Reproducible Public-Data Benchmark of Birthweight-for-Gestational-Age Deviations and Infant/Fetal Mortality Risk: A Cross-Sectional Ecological Analysis of United States Vital Statistics, 2016–2024

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Abstract

Background: Previous research has identified birthweight-for-gestational-age, small-for-gestational-age, and large-for-gestational-age categories as risk factors for perinatal outcomes; however, the majority of published reports that examine these categories have relied on aggregated data exported from administrative databases rather than on record-level public-use files that could be made available through open science. In order to identify fetal growth deviations, this study describes a method to test the following question: Can record-level public-use files of birthweight from the National Center for Health Statistics be transformed into a reproducible national aggregate-cell benchmark that can be compared to metrics of birthweight for gestational age against the birth-cohort linked infant death and delivery-based fetal death data surfaces? The study is designed as a cross-sectional, ecological (aggregate-cell) analysis of routinely collected vital-statistics records rather than an individual-level cohort study.

Objective: Create a repeatable public-data aggregate-cell benchmark using aggregated data exported from CDC WONDER (CDC Wide-ranging Online Data for Epidemiologic Research) and the National Vital Statistics System (NVSS) natality records, birth-cohort linked infant-death files, fetal-death files, and fetal-growth charts from the INTERGROWTH-21st fetal-growth study and the World Health Organization (WHO), and fetal-growth calculator tables from the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD), as well as the cited fetal-growth and Binary Cellular Analysis literature.

Methods: Using standard file transfer protocol (FTP) and streaming, we downloaded and uncompressed fixed-width record-level public-use files in the United States to extract birthweight/sex/gestational-age data. We combined birth-cohort linked infant-death denominators and numerators (two years) to create birth-cohort linked infant-death aggregate cells, and we combined natality records into singleton gestational-age, birthweight, and sex cells. In order to construct national live-birth, birth-cohort linked infant-death, period-linked infant-death sensitivity, and delivery-based fetal-death surfaces, we also extracted aggregated natality denominators and fetal-death numerators. Using gestational age in a log-cubic model, we fitted the primary sex-specific expected birthweight model; as a demonstration of how the benchmark adjudicates candidate growth equations, we also fitted a proxy equation derived from a Binary Cellular Analysis (BCA) framework proposed by Michaelson and coauthors using a dimensionless biomass proxy, $N^* = W/3500\,\mathrm{g}$, where W is birthweight in grams. For every model, residuals in gestational age and birthweight aggregate cells were computed. The birthweight percentile risk zones were contrasted with the percentile risk zones obtained from residuals. We conducted temporal out-of-sample, leave-one-year-out, and in-sample benchmark testing. Additionally, we compared the U.S. birthweight for gestational-age surface to international fetal growth references and standards, such as WHO, INTERGROWTH,

and the NICHD standards, which are tailored for each mother and child. To audit the source data and outcomes, we employed data-flow diagrams, hashes, count flows, fixed-width field maps, and source data manifests. The study is reported following the STROBE and RECORD guidelines for observational studies of routinely collected data.

Results: 33,589,282 singleton natality records, 29,633,197 period-linked linked birth-cohort infant death denominators, 163,804 period-linked linked infant death numerators, 29,950,846 cohort birth-cohort linked birth denominators, 143,079 cohort filtered linked infant death numerators, and 498,185 fetal death records were extracted from all sources. Cohort-linked infant mortality was 3.445 times higher (95% CI 3.395–3.495) and fetal mortality was 5.213 times higher (95% CI 5.145–5.282) in the below-p10 cell compared to the p10–p90 gestational age/birthweight cell. Endpoint asymmetry was clearly evident in the above-p90 cells, and fetal death was significantly increased (RR 1.315), although newborn cohort mortality was only marginally higher (RR 1.025); the fetal-versus-infant ratio of rate ratios above p90 was 1.282 (95% CI 1.241–1.325, $p<0.001$). In terms of temporal validation of the candidate-metric demonstration, the BCA residual remained within flexible birthweight-surface benchmarks while also closely matching the log-residual model for cohort infant mortality (AUC 0.777 versus 0.777) and fetus mortality (AUC 0.879 versus 0.879). Congenital anomalies accounted for 56.8% of the excess mortality among infants in the below-p10 category, according to the cause-specific cause-group decomposition, while the cause groups haematologic, hydrops, and infant-of-diabetic-mother (fetal cause group rate ratio: 7.18) accounted for the majority of the excess mortality among infants in the above-p90 category. Rather than growth-zone distribution disparities, within-growth-zone variations in mortality risk accounted for 84.3% of the difference in infant mortality between the Black and White cohorts (rate ratio 1.99), according to the Kitagawa decomposition. The birthweight surfaces defined by population, WHO, INTERGROWTH-21st, and customized criteria discriminated mortality nearly identically, and the growth–mortality connection stayed constant throughout the COVID-19 era.

Conclusions: In order to test birthweight-for-gestational-age (fetal growth deviations) metrics for results related to the birth-cohort connected infant death, period-linked infant death sensitivity, and fetal death data surfaces in the United States, we have developed a replicable public-data aggregate-cell benchmark. The study's most obvious new finding was that the risk of excessive mortality by gestational age/birthweight cell varied by endpoint. Specifically, infants in the above-p90 growth-zone cell were more strongly associated with the delivery-based fetal mortality outcome than the birth-cohort linked infant death outcome. When compared to the standard birth-cohort log-cubic residual, which it closely matches in out-of-sample validation benchmarks, the BCA fetal-biomass proxy may be viewed as a compact benchmark-equivalent residual, illustrating how the framework adjudicates candidate growth metrics. Because the benchmark is constructed from aggregate cells, these findings are population-level descriptive associations; they do not support individual-level prediction or causal inference.

Plain-Language Summary

This study produces maps that illustrate the changes in a baby's mortality risk based on gestational age, gender, and birth weight using unfiltered and publicly available birth and death data in the United States. Deliveries smaller than expected for their gestational age experienced higher incidences of newborn death and fetal death compared to those in the middle weight range for their gestational age. The risk of newborn death in infants who are heavier than their gestational age was found to be only slightly higher, while the risk of fetal death was considerably higher. Furthermore, a simple growth equation based on biological principles was applied to analyze the same risk in years of previously collected data and the performance was comparable to that of the traditional curve.

What This Study Adds

- It adopts the publicly available NCHS standard data obtained from the natality and fetal death sources, rather than using similar summary tables from available census sources.

- This model generates various accounts for the number of fetal deaths corresponding to the time of birth as well as those dependent upon the specific death rates and birth cohorts.
- This model utilizes empirical p10/p90 data as well as small spread log-cubic residuals and modelled birth weight data while using a variety of checks regarding time of validation methods such as cross-validation, partial validation and time-of-validation checks.
- As a worked example of candidate-metric adjudication, it finds a BCA residual proxy which closely follows the log-cubic residuals from the withheld data and that the risk of delivery-based fetal death is greater than the risk of cohort-based birth infant death, both at above-p90.
- It provides a potential cause-of-death explanation for the high-growth-zone signal showing that the below p10 infant mortality excess primarily involves congenital anomalies and that the above p90 is primarily hematologic/hydrops/infant of a diabetic mother.
- It applies a Kitagawa decomposition which shows that the majority of the Black-White perinatal mortality gap exists within the growth zones and not between them.
- It shows that the growth-mortality surface remained unchanged with the COVID-19 natural experiment, and that the WHO, INTERGROWTH-21st, and custom growth standards differentiated perinatal mortality almost the same.

1 Introduction

Fetal growth monitoring represents a relevant problem for health practitioners and officials alike. Even if universal models let comparisons be made, local criteria seem fit to describe the population undergoing measurement more accurately while outcome-based models are believed to suit pregnancies at an increased risk much better. People are less concerned with the mathematical properties of the model curves than whether such deviations present actual differences in fetal and infant mortality rates.

SGA has a known association with negative perinatal outcomes [1–5]. The main challenge here is the reproducibility of methods and clarity of denominators. The publicly available NCHS data contain natality records, period-linked infant-death records, cohort-linked infant-death records, and fetal-death records. Each of these data sets represents a different aspect of the same problem. Consequently, one growth category may have different meanings, depending on the endpoint, e.g. fetal death at delivery, infant death after birth, or a period-based weighted sensitivity endpoint.

This research fills that gap by generating a publicly available data base derived from the records and using it to investigate four significant questions: the cause-of-death structure that is responsible for the mortality excess in the growth zone, the portion of the black-white perinatal mortality gap linked to growth-distributional risk and the share attributable to the risk within the zone, the impact of the COVID-19 crisis on the growth-mortality relationship, and the growth benchmark able to reflect mortality the best. Each surface of mortality includes the operation of a number of steps, ranging from extracting data from fixed-width files to building the final tables. The cause-of-death decomposition provides descriptive explanations to the above-p90 fetal signal that was primarily informative, whereas the disparity decomposition can be used for informing policy discussion. Using the BCA results as a worked demonstration of candidate-metric adjudication instead of making them the main claim is also a notable feature of this research. By BCA, we mean Binary Cellular Analysis framework. It was developed by Michaelson and his collaborators, is currently available only as preprints, and is applied in this study strictly as the source of a candidate biomass proxy equation for benchmarking, not as established or peer-reviewed biology [6, 7]. Throughout, the design is ecological: every association is estimated at the level of aggregate gestational-age/sex/year/weight-band cells, and no individual-level or causal interpretation is intended.

2 Data

Analysis results are dependent upon the files stemming from public sources mentioned in the project manifest. For public use, NCHS natality, linked birth/infant death, and fetal death are sourced from ZIP Files available [8–11]; WHO, INTERGROWTH-21st, and NICHD fetal growth standards are derived from publicly available data in spreadsheets, PDF files, and DOCX documents [12–15]. Mentioned only as references are Grantz and BCA manuscripts/preprints, while the relevant literature does not contain any data on the possible outcomes [6, 7, 16]. The outcomes described in source audit originate from public records or deterministic transformations of those records. Shown in Table 2, the consistency of records obtaining information from NCHS records are discussed via denominator concordance tests. For audit reasons, the whole source list as well as CDC WONDER, counts of the records is given in a supplementary material. Reflected through public records, the total number of records obtained filters to sources used. Table 1 displays the data according to distinct categories of rows: data from the source rows, rows with unique valid births, linked rows for different periods, linked data for the birth cohort for a user, rows for fetal deaths, the structured elements of the model, and the different elements from the comparison of the models. The terms used to explain the denominator can differ for the analysis of the birth cohort and fetal death; in particular, for the infant death analysis, denominator cells include the number of births for the studied cohort for the period from 2016 to 2023, which were used to calculate the value with sensitivity analysis and period weights. The analysis of fetal deaths is based on the data about births and fetal death for the period from 2016 to 2024; the parsed fetal-death source files extend back to 2014, as reflected in Table 1, but all fetal analyses are restricted to 2016–2024. Thus, the resulting analytic file consists of 19,166 records, providing 28,872,085 births and 143,077 infant deaths used in the infant death analysis, and 32,448,118 births, 179,027 cases of fetal deaths, and 32,627,145 fetal-denominator observations used in the fetus mortality analysis.

Table 1: Count-flow table separating records, rows, weighted deaths, model-ready cells, and model-comparison cells.

Stage	Years	Unit	Rows/cells	Natality	Linked denom	Infant num	Fetal deaths	Fetal denom
Raw source records	source-specific	records	63884468	33589282	29633197	163804.0	498185	
Singleton valid natality rows	2016-2024	rows	32448118	32448118				
Singleton valid linked denominator rows	2017-2024 period files	rows	28644762		28644762			
Singleton valid linked infant-death rows	2017-2024 period files	rows / weighted deaths	141354			142632.4		
Singleton valid cohort denominator rows	2016-2023 birth cohorts	rows	28872085		28872085			
Singleton valid cohort infant-death rows	2016-2023 birth cohorts	rows / unweighted deaths	143079			143079.0		
Singleton valid fetal-death rows	2014-2024	rows	222075				222075	

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Table 1: Count-flow table separating records, rows, weighted deaths, model-ready cells, and model-comparison cells.

Stage	Years	Unit	Rows/cells	Natality	Linked denom	Infant num	Fetal deaths	Fetal denom
Harmonized infant model-ready cells	2016-2023 birth cohorts	aggregate cells	15314		28872085	143077.0		
Harmonized fetal model-ready cells	2016-2024	aggregate cells	19101	32448118			179027	32627145
Infant model-comparison cells	2016-2023 cohorts, 24-42 weeks	aggregate cells	6111		28750979	99253.0		
Fetal model-comparison cells	2016-2024, 24-42 weeks	aggregate cells	7007				96602	32419577

Table 2: Linked denominator and natality singleton-count concordance.

Year	Births	Linked births	Difference	Rel diff (%)
2017	3727649	3728351	702	0.019
2018	3669736	3670407	671	0.018
2019	3628705	3629448	743	0.020
2020	3499833	3500729	896	0.026
2021	3547665	3549554	1889	0.053
2022	3553730	3555454	1724	0.049
2023	3487259	3488996	1737	0.050
2024	3519989	3521823	1834	0.052

3 Methods

3.1 Study Design, Setting, and Reporting Standards

This is a cross-sectional, ecological (aggregate-cell) analysis of routinely collected United States vital-statistics records for 2016–2024. The unit of analysis is the aggregate cell rather than the individual pregnancy, a design chosen so that every reported estimate remains reproducible from public files and stable under small-cell privacy limits. The study is reported in accordance with the STROBE and RECORD guidelines [17, 18]; the completed item-by-item checklist is provided in the supplement (S1 File).

3.2 Records, Eligibility, and Attrition

Our analysis was confined to singleton live births and fetuses that died but had usable sex, gestational age via obstetric estimate, and usable birthweight or delivery weight. We streamed the fixed-width records of file-level NCHS from the public ZIP files and performed a collapse into cells that are classified by sex, gestational age, year and 100-grams weight band. In this paper the term aggregate bin or aggregate cell denotes one unique combination of year, sex, completed weeks of gestation and 100-gram weight band. Table 1 reports the full record flow from raw source records through eligibility filtering to analytic cells, serving as the participant-flow summary for this record-based design. Records with unknown or missing sex, gestational age, or birthweight were excluded rather than imputed (a complete-case design

at the record level), and Table 1 quantifies these exclusions. No a priori sample-size calculation was performed; the study uses the complete census of all eligible national records for 2016–2024.

3.3 Endpoints and Linkage

The main infant endpoint relies on user-constructed birth-cohort linked files, built programmatically by the released scripts following the NCHS cohort-construction procedure: the cohort-year denominator file was combined with the numerator files from the year of birth and the following year, which included only valid newly-born infants. No step of this construction is performed by hand; the complete procedure is executed by the public code, so "user-constructed" refers to the NCHS convention that analysts must assemble cohort files from the released period files. The National Center for Health Statistics reports that birth-cohort files are considered to be applicable to tracking the status of infants born in a certain year and are more useful for complex analysis than other types of files and without corresponding period record weight RECWT [10, 19]. The only records that include period-weighted estimates of infant mortality are considered as sensitivity endpoints. By following the guidelines of NCHS, the birth-cohort denominator and numerator indicate individuals who are U.S. residents, while natality, period-linked, and fetal death surfaces show all cases when the death occurred in the U.S.; however, the percentage of residence-excluded deaths is around 0.2–0.3%, which does not significantly affect the obtained results on the growth zones.

The fetal death records have been matched to live births according to the same year, sex, gestational age, and weight bands. The NCHS fetal mortality figures conventionally relate to fetal deaths beyond 20 weeks per 1,000 live births plus fetal deaths beyond 20 weeks [11, 20]. The fetal tables presented here apply to gestation in the range of 24-42 weeks only and stress delivery-based, weight-band-specific mortality patterns. The denominator is different from the corresponding fetal survival calculation [21]. Several potential sources of bias attach to these linkages. The small fraction of infant-death records that NCHS cannot link to a birth certificate is absent from the cohort numerators, which slightly undercounts infant deaths; gestational age based on the obstetric estimate is subject to misclassification and digit preference; and the delivery-based fetal denominator counts deliveries rather than ongoing pregnancies, so it should not be read as a fetuses-at-risk hazard. These directions of bias are described rather than corrected, consistent with the descriptive design.

3.4 Expected-Size Models

Two expected size models were utilized in the study to estimate the national mean birthweight as a function of the gestational week from 22 to 42 weeks post-partum. The first of these was a compact sex-specific log-cubic model that is already known. The second model is a demonstration candidate: it used the BCA age-size equation, a compact proxy drawn from a preprint framework, to show how the benchmark adjudicates a new candidate growth equation against the empirical baseline. The BCA proxy is defined through the dimensionless biomass proxy $N^* = W/3500\,\mathrm{g}$; its mitotic-fraction equation, growth equation, integrated age-size form, and fitted parameters are given in the supplement. Each model was fitted separately for male and female births, and the integrated equation was inverted numerically to obtain BCA-expected weight at each completed gestational week.

3.5 Growth Zones and Cell Restrictions

We did a comparison based on the p10/p90 empirical zones, mortality models of aggregate cells and residual bins when it came to mortality surfaces of infants and fetuses. For fetal death analysis, empirical growth zones were obtained by the comparison of the midpoints of delivery weight bands with the sex- and gestational-age-specific p10 and p90 live-birth thresholds. The summaries of growth zone mortality based on bands were limited by the number of weeks of gestation together with cells that have their

denominators as positive. The models were restricted to 24-42 completed weeks while using only cells with denominator reaching not lower than 50. Also the models made sure to exclude cells that were internally inconsistent together with aggregated cells showing when weighted deaths exceeded their denominators.

3.6 Statistical Methods

The rates of death are based on models that employ grouped binomial generalized linear models. The response variable is the death proportion in each aggregate cell, with the cell denominator supplied as a binomial variance weight. It is the same as making a death modeling against non death in the cell. The generalized linear models used a binomial family with a logit link, estimated by maximum likelihood in Python 3.13 with statsmodels 0.14.6, pandas 2.3.3, and numpy 2.4.4; expected-size curves were fit with scipy 1.17.1. Formal comparisons between endpoints used grouped-binomial models with zone-by-endpoint interaction terms, and formal comparisons between validation AUCs used a cluster bootstrap that resamples validation aggregate cells with replacement (2,000 replicates), reporting percentile 95% confidence intervals and two-sided bootstrap p-values; both are reported in the supplement.

In the case of model comparison, the baseline which consisted of GA only had gestational age and gestational age squared. The models including p10/p90, residual, BCA residual, and birthweight-surface, had fetal/infant sex as a covariate in addition to gestational age and gestational age squared. The primary benchmark is deliberately unadjusted beyond gestational age and sex: jointly stratifying the aggregate cells by additional maternal covariates such as smoking, body-mass index, or pre-pregnancy conditions would multiply the number of cells and push most below privacy-stable sizes, so those covariates are reported as descriptive strata in the supplement, and race enters only the prespecified Kitagawa decomposition through race-stratified cells. Individual-level covariate adjustment requires record-level modeling outside this aggregate design and is identified as next-stage work. As a sensitivity analysis, the growth-zone models were additionally refit on race-stratified cells with and without race/Hispanic-origin covariates (supplement). To summarize in-sample model performance, Akaike information criterion (AIC), Brier score, weighted area under receiver operating characteristic curve (AUC) and deviance pseudo-R2 were used. As these grouped binomial models are fit with aggregate-cell variance weights, the reported AIC is a variance-weighted pseudo-likelihood used only for within-outcome model ranking, not as an absolute information criterion comparable between outcomes. Weighted AUC constructs positive event weight and negative non-event weight for each of the aggregate cells. Brier scores were calculated as denominator-weighted mean squared errors, and hence their absolute values are small, since fetal and infant deaths are rare.

The models on infants are validated for the period of 2016-2020 and their counterparts on fetuses for 2016-2021, with validation of the infant models for 2021-2023 and of the fetal ones for 2022-2024. The result of the validation is represented by Brier score (with denominator weighting), AUC (weighted), and validation negative log-likelihood (not AIC). One-year-out validation, the use of calibration deciles, and net-benefit decision-curve analyses are presented in the appendix. The abbreviations of the models include those with GA alone, GA + p10/p90, GA + log residual, GA + BCA residual, and GA + BW surface.

Confidence intervals for the rate ratio of empirical growth zones use the empirical p10–p90 zone as the point of comparison; 95% intervals use the normal approximation on the log rate ratio with standard error $\sqrt{1/d_1 - 1/n_1 + 1/d_0 - 1/n_0}$, where d and n denote zone deaths and denominators. Endpoint and period contrasts of these rate ratios use the delta-method ratio-of-rate-ratios with a z-test. Data on infant growth zones have been obtained through the mortality data linked to birth cohorts. Period-based weighted information and the check eliminating from the database the information on periods with weighted mortality that is greater than the corresponding mortality on the linked denominators are provided as supplementary information. As for the rates stratified based on various covariates, they are treated mostly as feasibility and sense-checking data provided as additional information rather than

being included into the main results. This data summarizes publicly available information for future adjustments.

3.7 Reporting, Sensitivity Analyses, and Reproducibility

This supplement provides the entire list of sources for events, a count of the records made, a map of all fields, and data tables for BCA proxy parameters, race-adjusted growth-zone sensitivity models, validation over time, validation of the effect of miscounting errors made in each year, simulation of data with one less year to analyze, summaries of the decisions made by the model, infant predictions made, and further details of the coding process of the events mentioned above [17, 18].

3.8 Ethics, Funding, Conflicts, and Data Availability

Publicly available anonymized data from vital statistics was used in this study, as well as publicly available fetal growth standards; as such, there was no need to obtain IRB approval. No funds from third parties were requested for this study. The author asserts that he does not have any competing interests. Morsal Mubarak was responsible for study conception, data collection, analysis preparation, coding, maintaining audit history and writing of the paper. All source datasets were publicly available. Analysis scripts, processed audit CSV files, data provenance audit logs, SHA-256 hashes, and commands used for data regeneration are located and publicly available at https://github.com/mubarsad012/public-use-fetal-growth-benchmark, and are permanently archived on Zenodo at https://doi.org/10.5281/zenodo.21894038 (concept DOI, which always resolves to the most recent archived release). An always-on interactive audit dashboard and calibrated risk calculator are served statically at https://mubarsad012.github.io/public-use-fetal-growth-benchmark/app/, alongside the full static audit portal at https://mubarsad012.github.io/public-use-fetal-growth-benchmark/; both are hosted on GitHub Pages, are built deterministically from the released audit tables, and load with no server start-up or idling. A live mirror of the dashboard runs at https://public-use-fetal-growth-benchmark-mchf4ynlunqxyguja4viy3.streamlit.app/; it is hosted on Streamlit Community Cloud, which idles inactive applications, so a scheduled keep-alive workflow in the repository keeps the mirror responsive. All underlying outcome data are de-identified public-use NCHS files and CDC WONDER exports, as described in the Data section.

4 Results

The record-degree extraction comprised the same number of 33,589,282 records of birth and carried out the extraction of 29,633,197 denominator records linked over time, 163,804 infant mortality records linked over the same period, 29,950,846 denominator cohort records, 143,079 infant mortality records filtered for cohort effects, and 498,185 records of fetal mortality. Table 1 demonstrates the movement of the source records into valid singleton rows, counts of neonatal deaths in a cohort, cells converted for models, and cells of model comparison. Figure 2 illustrates the density of singleton live births obtained through record-level NCHS extraction.

4.1 Cause-Specific Architecture of Growth-Zone Mortality

By linking each infant and fetal death with a cause (NCHS 130 causes infant recodes and fetal initiating-cause codes) [10, 22], and putting together the dependent causes of death in the growth zone revealed the different mechanisms on both ends of the tail. For instance, below empirical 10th centile, malformations are the leading cause of the cohort infant mortality (56.8% of the below p10 mortality; cause-group rate-ratio of 7.14), a pattern consistent with many of the deaths of small-for-gestational-age infants carrying lethal-malformation codes, rather than growth-restriction-related codes alone. At the same

time, in the case of fetal death below p10, malformations had the highest within-zone rate ratio (11.66) while placental, cord and membrane cause gave the biggest interpretable share of the dependent deaths (20.6%). Moreover, it is noteworthy that beyond the empirical 90th percentile, the picture is reversed as the primary causes of deaths now become hematology, hydrops, and diabetic mother, which leads to a value of fetal rate-group ratio equal to 7.18 (fetal deaths attributed to the above-mentioned causes account for 73.5% of all deaths above p90) and a cohort infant death rate equal to 2.73, which suggests that the discussed fetal signal is concentrated in hydrops-fetalis- and maternal-diabetes-related pathways rather than in uniform risk among heavier babies; as an aggregate-cell decomposition, this is a descriptive association, not an individual-level causal mechanism. All cause-group shares are death-certificate code shares, and their interpretation is bounded by coding quality: unspecified/ill-defined causes form the single largest cause group for below-p10 fetal deaths (supplement), so fetal cause-specific patterns in particular should be read as coding distributions rather than verified clinical causes. Due to the low absolute number of the excess above p90, Table 3 and Figure 1 show the percentages at its location.

Table 3: Decomposition of the below-p10 and above-p90 perinatal mortality excess into cause groups. The excess rate is the zone-specific minus reference (p10-p90) cause-group rate; pct_of_zone_excess is each group's share of the total zone excess.

endpoint	growth zone	cause group	zone rate per 1000	reference rate per 1000	excess rate per 1000	rate ratio vs reference	pct of zone excess
Cohort infant death	Below empirical p10	Congenital anomalies	4.93	0.69	4.24	7.14	56.81
Cohort infant death	Below empirical p10	Placental/cord/membrane	0.10	0.06	0.03	1.54	0.45
Cohort infant death	Below empirical p10	Asphyxia/hypoxia	0.12	0.06	0.06	2.12	0.83
Cohort infant death	Below empirical p10	Maternal complications	0.16	0.08	0.09	2.14	1.18
Cohort infant death	Below empirical p10	Prematurity/growth-related	0.51	0.08	0.42	6.03	5.64
Cohort infant death	Below empirical p10	Perinatal respiratory	0.48	0.16	0.32	2.97	4.26
Cohort infant death	Below empirical p10	Infection	0.55	0.26	0.29	2.14	3.94
Cohort infant death	Below empirical p10	Haematologic/hydrops/IDM	0.27	0.15	0.12	1.78	1.60
Cohort infant death	Below empirical p10	SIDS/SUID/ill-defined	1.30	0.63	0.67	2.07	9.00
Cohort infant death	Below empirical p10	Other perinatal	0.41	0.17	0.24	2.39	3.19
Cohort infant death	Below empirical p10	Other/residual	1.69	0.71	0.98	2.38	13.10
Cohort infant death	Above empirical p90	Congenital anomalies	0.88	0.69	0.19	1.27	244.84
Cohort infant death	Above empirical p90	Placental/cord/membrane	0.06	0.06	-0.00	0.98	-1.88
Cohort infant death	Above empirical p90	Asphyxia/hypoxia	0.08	0.06	0.02	1.38	27.86
Cohort infant death	Above empirical p90	Maternal complications	0.08	0.08	0.01	1.10	9.60
Cohort infant death	Above empirical p90	Prematurity/growth-related	0.05	0.08	-0.03	0.64	-39.56
Cohort infant death	Above empirical p90	Perinatal respiratory	0.12	0.16	-0.04	0.75	-52.52
Cohort infant death	Above empirical p90	Infection	0.17	0.26	-0.09	0.67	-111.46
Cohort infant death	Above empirical p90	Haematologic/hydrops/IDM	0.42	0.15	0.27	2.73	347.48
Cohort infant death	Above empirical p90	SIDS/SUID/ill-defined	0.42	0.63	-0.21	0.67	-274.07

endpoint	growth zone	cause group	zone rate per 1000	reference rate per 1000	excess rate per 1000	rate ratio vs reference	pct of zone excess
Cohort infant death	Above empirical p90	Other perinatal	0.23	0.17	0.06	1.32	72.54
Cohort infant death	Above empirical p90	Other/residual	0.61	0.71	-0.09	0.87	-122.83
Fetal death	Below empirical p10	Congenital anomalies	1.86	0.16	1.70	11.66	16.31
Fetal death	Below empirical p10	Placental/cord/membrane	2.89	0.75	2.14	3.85	20.57
Fetal death	Below empirical p10	Asphyxia/hypoxia	0.01	0.00	0.01	3.70	0.08
Fetal death	Below empirical p10	Maternal complications	2.21	0.40	1.81	5.48	17.36
Fetal death	Below empirical p10	Prematurity/growth-related	0.46	0.03	0.43	15.56	4.17
Fetal death	Below empirical p10	Perinatal respiratory	0.01	0.00	0.01	6.11	0.06
Fetal death	Below empirical p10	Infection	0.01	0.00	0.01	5.05	0.09
Fetal death	Below empirical p10	Haematologic/hydrops/IDM	0.30	0.09	0.21	3.23	1.98
Fetal death	Below empirical p10	Unspecified/ill-defined	5.04	1.01	4.03	5.00	38.70
Fetal death	Below empirical p10	Other perinatal	0.04	0.01	0.03	4.40	0.29
Fetal death	Below empirical p10	Other/residual	0.05	0.01	0.04	4.60	0.39
Fetal death	Above empirical p90	Congenital anomalies	0.26	0.16	0.10	1.64	13.21
Fetal death	Above empirical p90	Placental/cord/membrane	0.57	0.75	-0.18	0.76	-23.68
Fetal death	Above empirical p90	Asphyxia/hypoxia	0.00	0.00	0.00	1.44	0.18
Fetal death	Above empirical p90	Maternal complications	0.51	0.40	0.10	1.25	13.10
Fetal death	Above empirical p90	Prematurity/growth-related	0.03	0.03	-0.00	0.99	-0.03
Fetal death	Above empirical p90	Perinatal respiratory	0.00	0.00	0.00	3.13	0.33
Fetal death	Above empirical p90	Infection	0.00	0.00	0.00	1.86	0.25
Fetal death	Above empirical p90	Haematologic/hydrops/IDM	0.66	0.09	0.57	7.18	73.53
Fetal death	Above empirical p90	Unspecified/ill-defined	1.15	1.01	0.14	1.14	18.44
Fetal death	Above empirical p90	Other perinatal	0.01	0.01	0.01	1.60	0.69
Fetal death	Above empirical p90	Other/residual	0.04	0.01	0.03	3.71	3.98

4.2 Decomposing the Black–White Perinatal Mortality Gap

A Kitagawa decomposition [23, 24] divided the gap in perinatal mortality between non-Hispanic Blacks and non-Hispanic Whites into a component which is growth-distribution (other ways of distribution in different growth zones) and a component which is within-zone-zone-specific risk (Table 4). While studying infant deaths, it was found out that the Black-White rate ratio is 1.99 (the gap is 3.18 per 1,000) and 84.3% of this absolute gap is accounted for by the mortality rates within the same growth zone while only 15.7% belongs to differences in distribution among zones. Going deeper into fetal deaths' comparison, the mortality rates were found out to be 2.09 (whose gap is 3.81 per 1,000) where 78.5% of

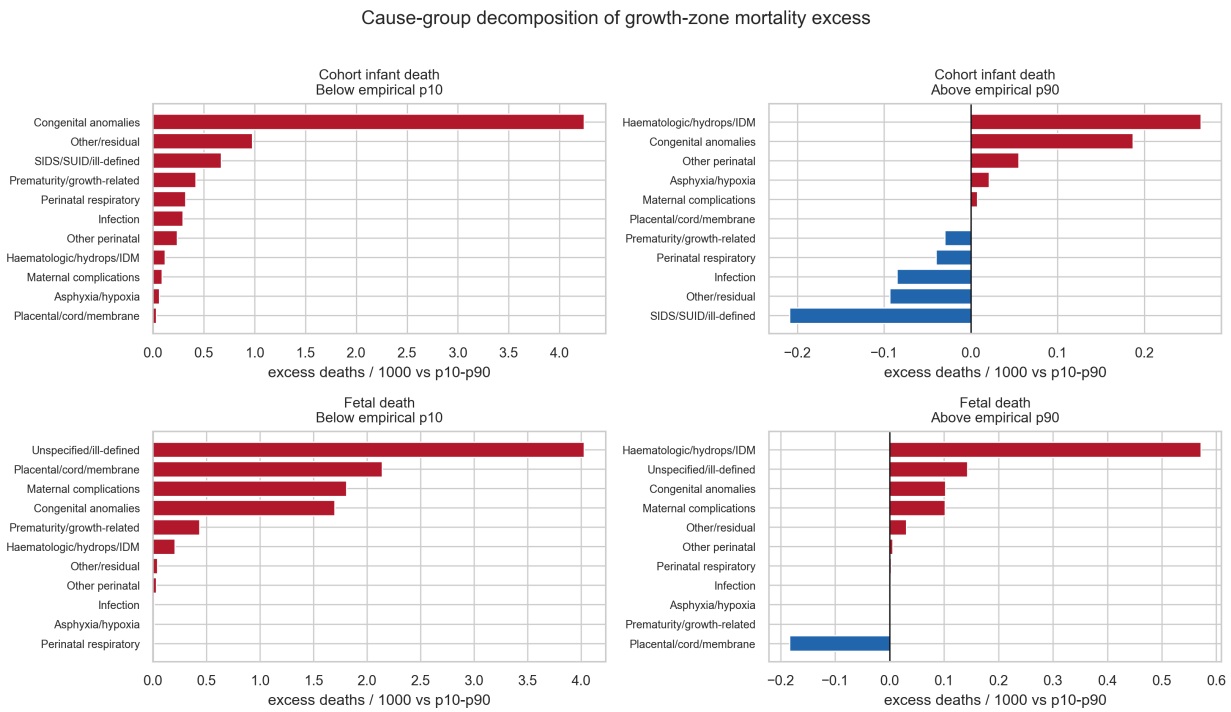


Figure 1: Cause-group decomposition of the below-p10 and above-p90 perinatal mortality excess (per 1,000) relative to the empirical p10–p90 reference zone, weeks 24–42, for cohort infant and fetal death.

the gap is attributed to mortality rates within the same zone and 21.5% to differences in distribution. Thus, in this descriptive accounting, the main reason for the disparity is the higher mortality within the same growth zone rather than a different distribution across zones: even though Black infants have a somewhat higher chance of being growth-restricted, the drivers of the higher within-zone mortality cannot be identified from these aggregate data. Therefore, race and ethnicities are viewed as social stratification rather than a biological explanation.

Table 4: Kitagawa decomposition of the non-Hispanic Black versus non-Hispanic White perinatal mortality gap into a growth-distribution component (different distribution across empirical growth zones) and a zone-specific-risk component (different mortality within the same zone), gestational weeks 24-42.

endpoint	white overall per 1000	black overall per 1000	absolute gap per 1000	rate ratio black white	distribution component per 1000	rate component per 1000	distribution pct of gap	rate pct of gap
Cohort infant death	3.21	6.38	3.18	1.99	0.50	2.68	15.65	84.35
Fetal death	3.49	7.30	3.81	2.09	0.82	2.99	21.50	78.50

4.3 Stability of the Growth–Mortality Relationship Across the COVID-19 Shock

The pandemic situation was analyzed as a natural experiment; hence, the growth-zone rate ratio was determined for the pre-pandemic period (2016-2019), pandemic period (2020-2021), and post-pandemic period. The below-p10 infant rate ratio was almost unchanged (for pandemic vs pre-pandemic; ratio-of-rate-ratios = 1.027, p=0.14), while a GA with log-residual model developed on the pre-pandemic data kept predicting its ability to discriminate between the pandemic and post-pandemic periods (weighted AUC of approx. 0.78 for cohort infants and approx. 0.88 for fetal death across periods). The only

visible change is a small reduction of the fetal below-p10 slope during and after the pandemic (ratio-of-rate-ratios 0.944 during and 0.933 after the pandemic, both p<0.001). The growth–mortality surface appeared stable across this external shock to obstetrical services and monitoring of stillbirths, with one small fetal exception.

4.4 Growth-Standard Adjudication by Mortality Discrimination

The effectiveness of competing growth standards was judged by their ability to discriminate perinatal mortality from out-of-sample cases using a shared grouped-binomial model. Validation weighted AUCs for infant mortality were similar amongst all population (empirical, sex-specific), WHO, or INTERGROWTH-21st references (all yielding a value of 0.781), while for fetal mortality the values were still very close (0.886, 0.886 and 0.885). The custom standard taking sex and ethnicity into consideration (GROW-type [25]) did not add any value regarding discrimination in relation to the reference population standard (infant mortality of 0.762 vs 0.766; fetal mortality of 0.876 vs 0.876). This customized comparator was evaluated because customized charts remain in clinical use and in the published literature, not as an endorsement of race-based customization; race enters it strictly as a social stratification variable. Its failure to improve discrimination is consistent with, and lends empirical support to, the ongoing move away from race-based growth customization in obstetric practice. Thus, taking into account the national level of mortality discrimination, the content of a particular growth standard made little difference; discrimination relies on the severity of the size-for-gestational-age deviation rather than on the reference against which this deviation is calculated. The similar nature of this "independence" is analyzed in the BCA proxy section below. The NICHD reference tables analyzed in this study are biometry focused and are not comparable in terms of birthweight, thus in the case of custom comparison median birthweights according to the empirical ethnicity factor were utilized.

4.5 Robustness Across the Specification Multiverse

The findings from the headline growth-zone were subjected to rigorous preplanned analysis across a total of 200 specifications which covered gestational-age duration, box weight range (100 to 500g), lowest available cell denominator and definition of fetal denominator. The result of the below p10 ratio showed a value greater than 2 for both of the endpoints across each of the specifications; it was also observed that the below p10 fetal ratio was greater than below p10 ratio of infants in each of the analyzed specifications as well. Furthermore, above p90 fetal rate ratio (i.e., fetal rate ratio being higher than infant rate ratio) was achieved in all tested and monitored specifications; however, the weaker condition where above p90 fetal rate ratio exceeds 1 was achieved in 97.5% specifications only. BCA proxy curve fitting was consistent regardless of the choice of reference biomass W_{REF} (the RMSE from the W_{REF} was below 0.001 g) indicating that reference serves as scaling convention and is not an independent defining factor for fitting. Detailed specifications of the curves are presented in supplementary material.

Table 5: Expected-size curve fit to record-level NCHS birthweight means. The BCA proxy equation uses N*=W/3500 g and a fitted gestational-age offset.

Model	Sex	GA pts	Births in fit	RMSE (g)	MAPE (%)	log R2
Log-cubic baseline	Female	21	15834450	12.1	0.338	0.999
BCA proxy equation	Female	21	15834450	13.7	0.422	0.997
Log-cubic baseline	Male	21	16586580	14.3	0.357	0.999
BCA proxy equation	Male	21	16586580	15.0	0.441	0.997

Table 6: Mortality rates and rate ratios by empirical growth zone.

Outcome	Growth zone	Deaths	Denominator	Rate/1000	RR (95% CI)
Cohort infant death	Above empirical p90	7350.0	2347654	3.131	1.025 (1.001-1.050)
Cohort infant death	Below empirical p10	23821.0	2263899	10.522	3.445 (3.395-3.495)
Cohort infant death	Empirical p10-p90	73931.0	24204382	3.054	1.000 (1.000-1.000)
Fetal death	Above empirical p90	8581.0	2641083	3.249	1.315 (1.285-1.344)
Fetal death	Below empirical p10	33106.0	2569428	12.885	5.213 (5.145-5.282)
Fetal death	Empirical p10-p90	67449.0	27289296	2.472	1.000 (1.000-1.000)

Table 7: In-sample aggregate-cell mortality model comparison. GA-only includes gestational age and gestational age squared; all non-GA-only rows also include sex. Cohort infant models use unweighted birth-cohort linked deaths; period-linked weighted models are reported only as sensitivity checks. Weighted AUC reconstructs positive event weights and negative non-event weights from each aggregate cell; full formulas are described in Methods. Cohort infant models used 6,111 cells, 99,253.0 deaths, and 28,750,979 denominator observations; Fetal models used 7,007 cells, 96,602.0 deaths, and 32,419,577 denominator observations.

Outcome	Model	AIC	Brier	Weighted AUC	Pseudo R2
Cohort infant death	GA only	959671.6	3.48e-05	0.747	0.816
Cohort infant death	GA + p10/p90	942163.4	2.26e-05	0.772	0.897
Cohort infant death	GA + log residual	936381.2	2.62e-05	0.780	0.924
Cohort infant death	GA + BCA residual	935988.0	2.58e-05	0.780	0.926
Cohort infant death	GA + BW surface	931911.0	2.15e-05	0.782	0.945
Fetal death	GA only	815565.2	9.05e-05	0.868	0.854
Fetal death	GA + p10/p90	792927.0	4.49e-05	0.880	0.913
Fetal death	GA + log residual	780655.7	2.04e-05	0.884	0.946
Fetal death	GA + BCA residual	780570.6	1.93e-05	0.884	0.946
Fetal death	GA + BW surface	781015.2	4.12e-05	0.887	0.945

Table 8: Temporal out-of-sample benchmark. Cohort infant models validate on 2021-2023 cells and fetal models validate on 2022-2024 cells; validation -2LL is reported instead of AIC.

Outcome	Model	Train years	Validation years	Validation Brier	Validation AUC	Validation -2LL
Cohort infant death	GA only	2016-2020	2021-2023	3.45e-05	0.746	426984.1
Cohort infant death	GA + p10/p90	2016-2020	2021-2023	2.13e-05	0.769	420644.3
Cohort infant death	GA + log residual	2016-2020	2021-2023	2.70e-05	0.777	418700.8
Cohort infant death	GA + BCA residual	2016-2020	2021-2023	2.68e-05	0.777	418569.9
Cohort infant death	GA + BW surface	2016-2020	2021-2023	2.13e-05	0.779	417171.0
Fetal death	GA only	2016-2021	2022-2024	7.27e-05	0.864	311840.8
Fetal death	GA + p10/p90	2016-2021	2022-2024	3.98e-05	0.875	305936.6
Fetal death	GA + log residual	2016-2021	2022-2024	2.43e-05	0.879	302707.0
Fetal death	GA + BCA residual	2016-2021	2022-2024	2.33e-05	0.879	302756.0
Fetal death	GA + BW surface	2016-2021	2022-2024	3.53e-05	0.882	302285.4

4.6 Primary Mortality Gradients (Internal-Validity Foundation)

Below-p10 cells demonstrated the most significant gradient of direct mortality, being associated with infant mortality rates 3.445 times higher than the respective p10–p90 reference range, as well as fetal

mortality 5.213 times higher than the benchmark (see Table 6 and Figure 5). This statement agrees with results of previous studies on SGA. Here, the role of this aspect is to provide a basis for validation of the parsing of raw data files, the datasets used in the study, and the process of creation of aggregated cells that reflect national-level mortality statistics prior to more detailed comparisons of indicator endpoints. Above-p90 cells have more pronounced endpoint observations where cohort infant mortality is only slightly higher than the threshold (RR 1.025) while fetal mortality shows a much larger increase (RR 1.315). A delta-method test of the fetal-versus-infant ratio of rate ratios confirmed the asymmetry formally: 1.282 (95% CI 1.241–1.325, $p<0.001$) above p90 and 1.513 (95% CI 1.484–1.543, $p<0.001$) below p10; a gestational-age-adjusted grouped-binomial model with zone-by-endpoint interaction terms gave concordant interaction estimates (1.281 above p90 and 1.318 below p10, both $p<0.001$) (supplement). In a race-stratified sensitivity analysis, adding race/Hispanic-origin covariates to the grouped-binomial zone models changed the gestational-age- and sex-conditional zone coefficients by less than 2% (below-p10: cohort infant 3.12 unadjusted versus 3.07 race-adjusted; fetal 3.27 versus 3.23), indicating that the growth-zone gradients are not materially confounded by race composition (supplement). This asymmetry is another clear signal from the benchmark, because a high-weight-for-gestational-age cell does not carry the same mortality risk for birth-cohort infant death that it does for delivery-based fetal death.

4.7 Candidate-Metric Demonstration: BCA Proxy Benchmarking and Validation

The close agreement of the expected-size curves (Table 5) shows that the low-parameter curves are a good representation of the national mean birthweight curve. Our validation question was whether deviations from these curves could group the aggregate cells into groups of mortality risk that differ from each other. In the comparison of the aggregate-cell models within samples (Table 7), the BCA proxy model behaved in a way that was similar to the log-cubic model but was still close to the flexible birthweight surface. The temporal validation confirmed this result (Table 8): BCA and log residual models achieved practically the same results on weighted AUC for cohort infant deaths (0.777 and 0.777) or fetal deaths (0.879 and 0.879) respectively. A cell-level bootstrap of the validation AUC difference showed that the log-residual model was nominally higher by less than 0.0002 in both endpoints (95% CI -0.00018 to -0.00002 for cohort infant death and -0.00023 to -0.00009 for fetal death), a difference that is statistically detectable but practically negligible (supplement). It is an important fact because it allows using this compact biologically inspired model, making the evaluation against a purely empirical model with the same national public data source transparent. This section functions as the benchmark's worked example: any candidate growth equation can be adjudicated in the same way, and the paper's substantive findings do not depend on the candidate evaluated here.

4.8 Calibration, Decision-Curve, and Descriptive Stratification Signals

In held-out validation, the highest predicted-risk decile of the BCA residual model contained 52.6% of cohort infant deaths and 73.7% of fetal deaths. The tables show decision-curve results with positive net benefit for the BCA residual and birthweight-surface models corresponding to 5 and 10 deaths per 1,000 thresholds across several flexible model families. From the tables, one can conclude about the effects of social stratification. For instance, the number of deaths of cohort infant children of Non-Hispanic Black ethnicity was 9.489 per 1,000 births compared to 4.013 Non-Hispanic Whites (RR = 2.365); and the number of fetal deaths among Non-Hispanic Blacks was 10.215 per 1,000 observations compared to 4.525 among non-Hispanic Whites (RR = 2.257). Race and ethnicity are viewed here as a socio-clinical stratification variable and not genetic explanation.

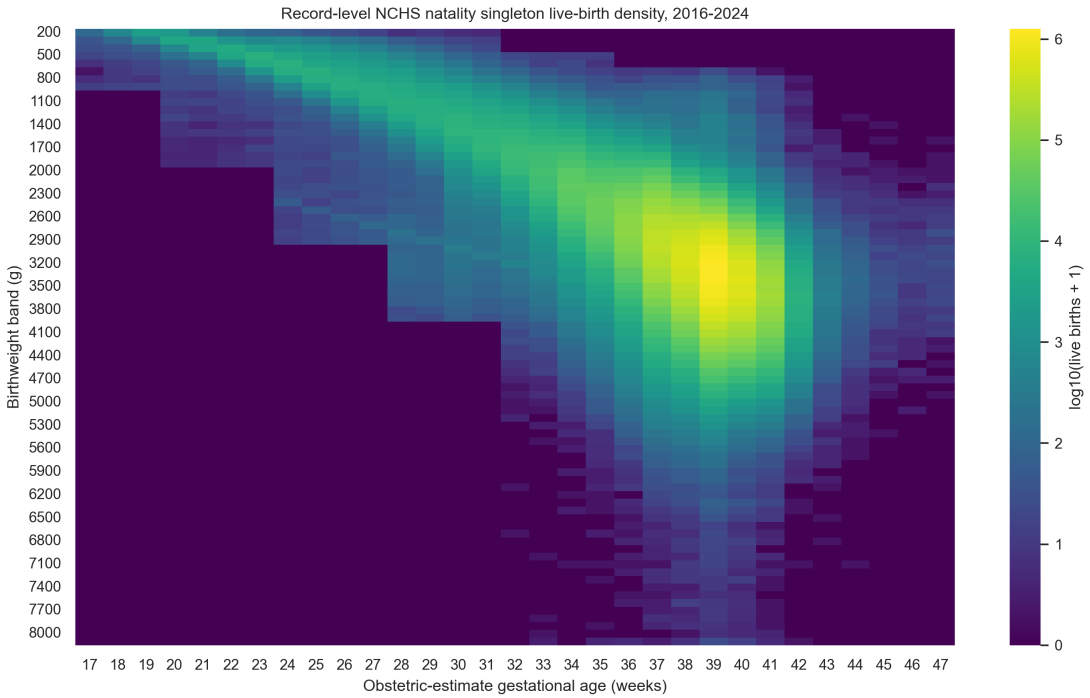


Figure 2: Record-level singleton live-birth density by gestational age and birthweight band. The color scale is $\log_{10}(\text{live births} + 1)$.

4.9 Outcome Surfaces

The surfaces for infant deaths and fetal deaths based on birth cohort connection (Figures 3 and 4) indicate that the risk is structured in a combined manner with the factors of gestational age and the weight of infants. Due to the nature of public aggregate data, which do not provide information about those pregnancies taking place in every gestational week, it is essential to treat the fetal death surface as the mortality surface based on delivery rather than the fetal survival hazard. The denominator of the fetal death indicator is an estimated aggregate one based on the number of fetal deaths plus the number of live births in the same year, sex, and age of infants.

5 Discussion

According to this research, the NCHS raw statistics regarding the public can be transformed into an open benchmark at a national level, which would allow fetal growth deviation metrics to be estimated easily. Known values of mortality below a certain limit can serve as an internal validity measure since the gained data can contain verified information about the considered risk of perinatal death.

The vital information is above the established threshold. In fact, infants that were large for their gestational age experienced only a slight increase in infant mortality at birth but a more visible increase in the fetal mortality rate. The aforementioned suggests a need to reconsider the way that the related data is read since the high weight of the child cannot be regarded as a universally applicable neutral measure.

Analyzing the specific causes is more than simply describing the zones of growth. The fact that congenital defects prevail in the group of infants born with a low body weight should not lead us to the conclusion that low birth weight is the only reason behind the excess infant mortality occurring in these infants. The reasons for the large number of infant deaths in the group of infants born with a

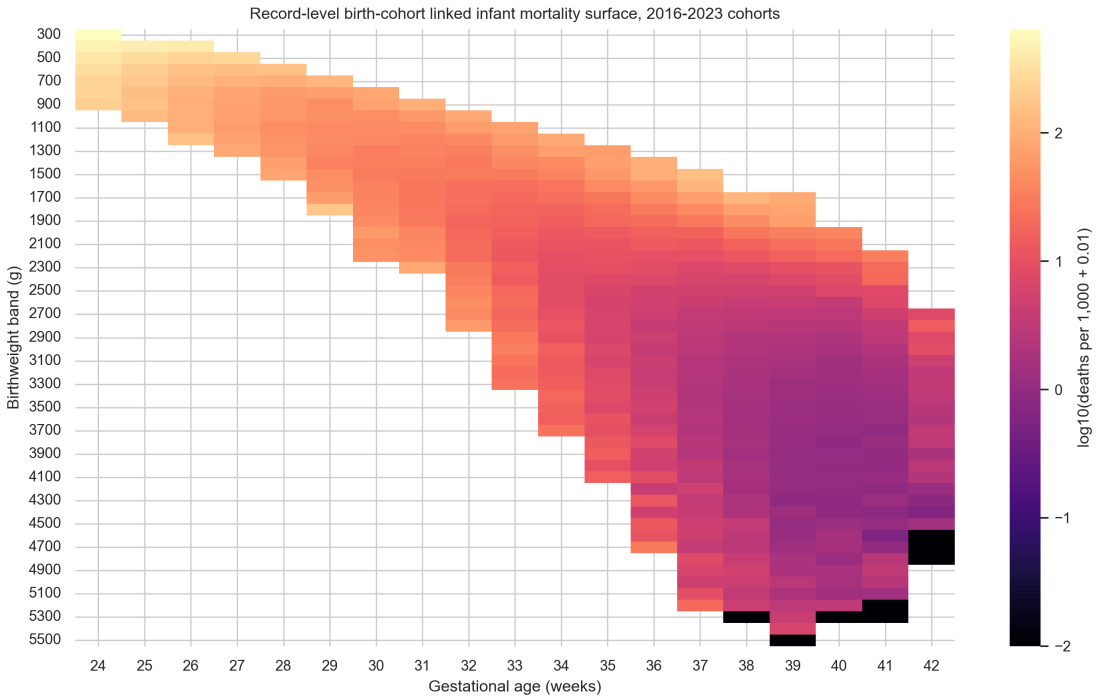


Figure 3: Record-level birth-cohort linked infant mortality surface, 2016–2023 cohorts. The color scale is $\log_{10}(\text{infant deaths per } 1,000 + 0.01)$.

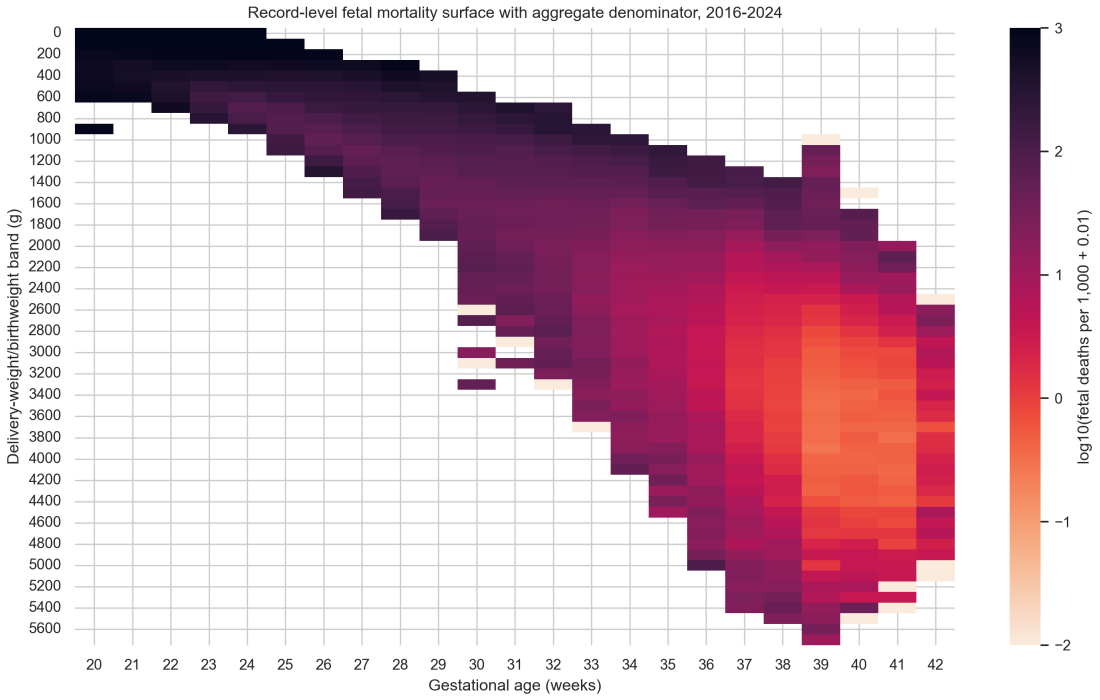


Figure 4: Fetal mortality surface using fetal deaths divided by fetal deaths plus matched live births, 2016–2024. The color scale is $\log_{10}(\text{fetal deaths per } 1,000 + 0.01)$.

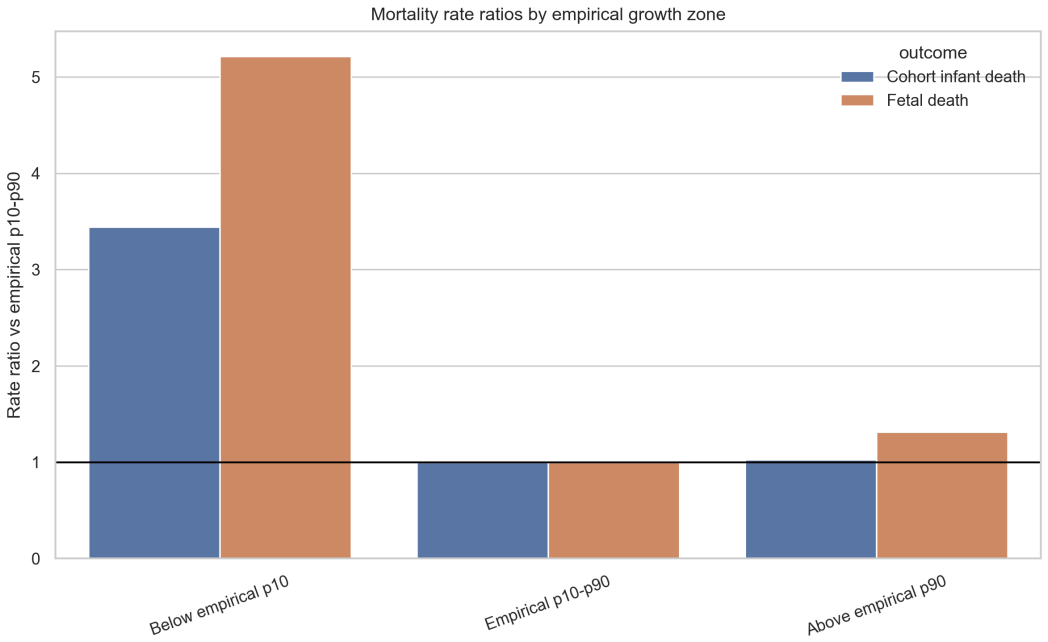


Figure 5: Mortality rate ratios by empirical growth zone, with p10–p90 as the reference zone. Ninety-five percent confidence intervals are reported in Table 6.

high weight are that the major causes of mortality among infants in this category are blood diseases, hydrops and the infant of diabetic mother.

The conclusion that can be derived from the study is that despite the fact that a large share of the excess rate of mortality among Black families in comparison with that among Whites originates from the problems within growth zones and not because the two races belong to different zones of growth, growth-distribution differences alone cannot account for the gap, and whether any particular intervention would narrow it cannot be inferred from these descriptive data.

Both the natural experiment conducted during the pandemic and the analysis of growth standards suggest that the relationship between weight and mortality did not look altered by the pandemic shock or by the choice of growth standard.

The comparison of the BCA proxies is more reflective of the benchmarks than of the major epidemiological assertion. The residuals from the proxy closely followed the log cubic residuals during the time validation and were still close to the flexible surface of the birth weight benchmark. This outcome might best be viewed as the equivalence of the benchmarks: a brief expression derived from the biological growth model can be robustly compared to the current empirical ones using public national data. Because the BCA framework itself has not yet undergone peer review, it functions here purely as a demonstration candidate; the benchmark's value does not depend on the framework's biological validity.

The major application of the given framework is the methodological one. It allows for the creation of a unique way of verifying how the growth measures and denominators and the designs of time validations can be defined and calculated prior to their application in practical uses and explanations. In interpreting all of these results, the ecological design sets the boundary: associations estimated on aggregate cells need not hold at the individual level, and none of the estimates reported here adjust for individual-level maternal covariates such as age, parity, body-mass index, or comorbidities. The benchmark is therefore positioned as a reproducible, population-level reference and hypothesis-generating resource rather than a clinical prediction or causal-inference tool. Because the data are a census of United States births and

deaths for 2016–2024, the descriptive results generalize to that national population and period, but not necessarily to other settings or to individual pregnancies.

6 Limitations

The results must be viewed in the context of the public access to vital statistics. The BCA process is a method of public data approximation based on final weight at birth; it is not a literal measurement of cells or a biological study of growth, and it is not evidence of cellular mechanism. The underlying BCA framework is available only as preprints and has not undergone peer review; it is used strictly as the source of a candidate equation for demonstration. The BCA proxy is not shown to replace conventional standards clinically; final body weight replaces the growth-rate data here. The fetal mortality risk surface calculates risk based on delivery rather than prospective fetuses; moreover, it uses only cells with determined weight at delivery and gestational age regarding the data aggregation. The analysis of public data is restricted because of CDC small-cell suppression rules, which limit the granularity of the analysis. The cohort approach is to analyze data at a collective level rather than to define causal relationships between groups of variables. Because the design is ecological, associations observed across aggregate cells cannot be assumed to apply to individual pregnancies (the ecological fallacy), and confounding by individual-level maternal characteristics cannot be excluded; the record-level covariate stratification in the supplement is descriptive only. Cause-of-death analyses inherit the limitations of death-certificate coding: misclassification is possible, unspecified/ill-defined causes are common, and for below-p10 fetal deaths they are the single largest cause group, so cause-group shares describe coding patterns rather than confirmed clinical mechanisms.

7 Conclusion

Cells based on derived record-level public vital statistics data of the U.S. can be constructed to create an auditable fetal and birth-cohort linked infant mortality surface. This approach is able to reproduce the already known below-p10 mortality trend and detect an above-p90 specific endpoint-based asymmetry where fetal mortality exceeds that of the birth cohort infants. Small deviations from expected birthweight-for-gestational-age, including the BCA proxy residual, are computable within this benchmark and are associated with mortality outcomes.

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